

1. The figure shows univariable associations with weight loss from a semaglutide t in Nature. The dot and whisker plots show what information?

Fig. 5: Effect of semaglutide treatment or placebo on mean percentage change in body weight from baseline to week 104 by subgroups.

Estimate	95% CI	P-value
Overall	-4.51 (-4.75 to -4.27)	
Sex		
Female	-11.11 (-10.86 to -10.86)	<0.0001
Male	-7.50 (-7.76 to -7.23)	
Age (years)		
<45	-6.26 (-6.50 to -6.02)	0.0003
45 to <65	-6.06 (-6.34 to -5.77)	
65 to <75	-6.26 (-6.53 to -5.99)	
≥75	-7.43 (-8.37 to -6.49)	
BMI (kg/m²)		
<30	-7.22 (-7.50 to -6.93)	<0.0001
30 to <35	-6.79 (-6.95 to -6.62)	
35 to <40	-6.01 (-6.26 to -5.76)	
≥40	-5.22 (-5.50 to -4.93)	
Body weight (kg)		
<90	-6.02 (-6.30 to -5.74)	0.0003
90 to <100	-6.26 (-6.53 to -5.99)	
100 to <110	-6.01 (-6.26 to -5.76)	
≥110	-7.48 (-8.37 to -6.59)	

a) Median and standard error

b) Estimate and confidence interval ✓

c) Forest residuals

2. The figure shows univariable associations with weight loss from a semaglutide t in Nature. Which pair of variables might raise collinearity problems if both were inc model?

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a) Sex and drug treatment group

b) Sex and age group

c) BMI and body-weight ✓

3. Which regression assumption is NOT really checkable from a diagnostic plot alo

Regression Diagnostics

Eight assumptions behind your linear model, and how to check each one in R

The SUNSHINE checklist - fill each blank with the whole word			
S Sample size Do you have enough observations for your model?	U uncorrelated errors Are the residuals independent of each other?	N No multicollinearity Are the predictors too correlated with each other?	S Shape, i.e. linearity Is the relationship really a straight line?
H homogeneity of variance Does the spread of the residuals stay constant?	I influential points Is any single point pulling the fitted line too much?	N normality of residuals Do the prediction errors follow a bell curve?	E exogeneity Is the predictor independent of the error term?

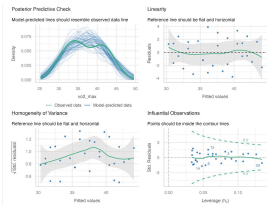
a) Exogeneity/Lack of confounding ✓

b) Normality of residuals

c) Shape (Linearity of the relationship)

d) Homogeneity of variance (Equal variance)

4. You run `check_model()` on a simple linear regression. The panels looks like the in How should you read it? (referring to image)



a) Linearity is violated

b) Linearity is acceptable ✓

5. You run `check_model()` on medical spending versus income. The Homogeneity of panel looks like the image. Which concern fits best? (referring to image)

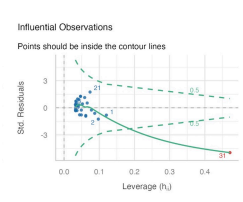


a) One observation has extreme leverage

b) Residual spread is not constant across fitted values ✓

c) Residuals are not normally distributed

6. You run `check_model()` on a BMI and cholesterol model. The Influential Observat looks like the image. What is the best next step? (referring to image)

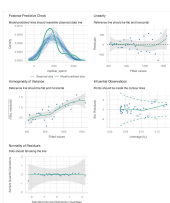


a) Investigate the unusual point, then consider a sensitivity analysis ✓

b) Delete the influential point

c) No particular action is needed as long as the overall R-squared remains high

7. You run `check_model` to assess normality of residuals and get the plot shown. W is correct?



a) Residuals are roughly normal and do not merit concern ✓

b) Residuals are worryingly skewed, in particular on the left hand side

